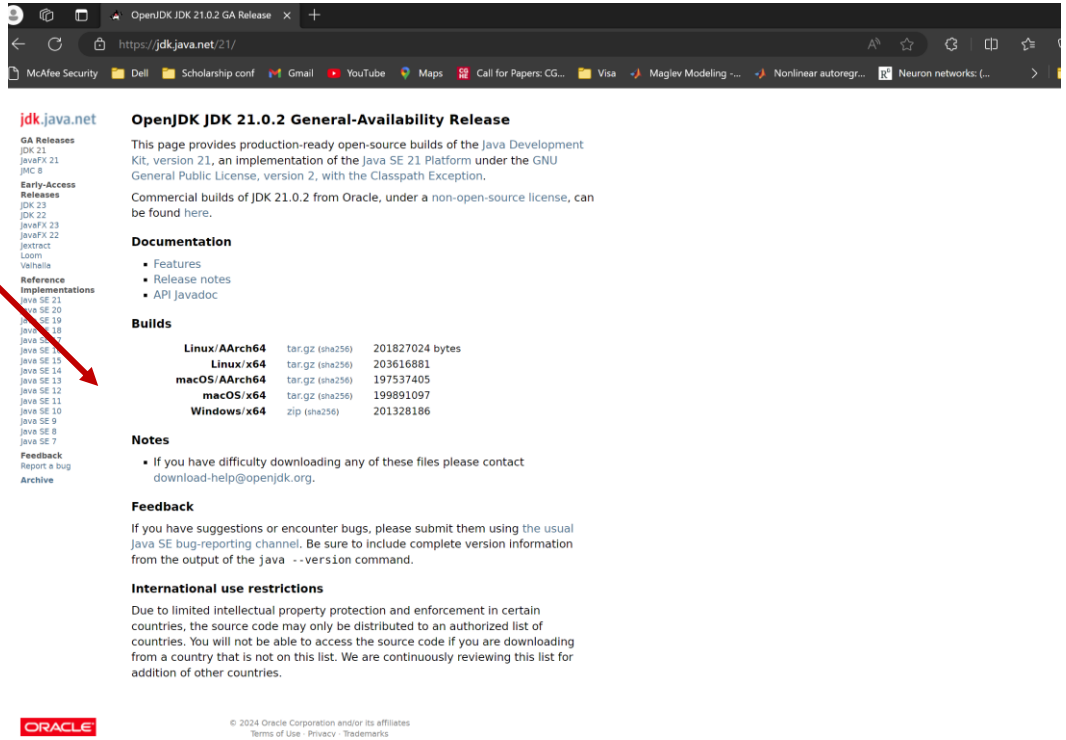
Installing the JDK & Essential Command Line Guide

IMPORTANT:

- It is assumed that you have access to a laptop (or desktop) computer.
  - If you do not have access to a computer, please email h.akter@qmul.ac.uk from one of your student email accounts (QMUL or BUPT) as soon as possible.
  - Otherwise, please try following all the instructions in this document as soon as possible; in particular, the information on pages *4-9* will help you complete the exercises.
  - Most of these slides assume also the use of a Windows OS computer.

Download and install OpenJDK 19

- Go to https://jdk.java.net/21/, select the Operating System (OS) of your computer and click on the appropriate link to download the software



- It is assumed that most students have access to a Windows OS computer (*); please follow the instructions here to install the software and configure the 'environment variables' on your computer.

(*) This is also what would be installed in the computing labs at BUPT.

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Checking the Configuration Settings on a Windows computer

- Start a **Command Prompt** window (also known as the **`cmd`** window)
- Type **`set`** → a detailed description of your OS environment settings will show up

```
Command Prompt
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\paula>set
```

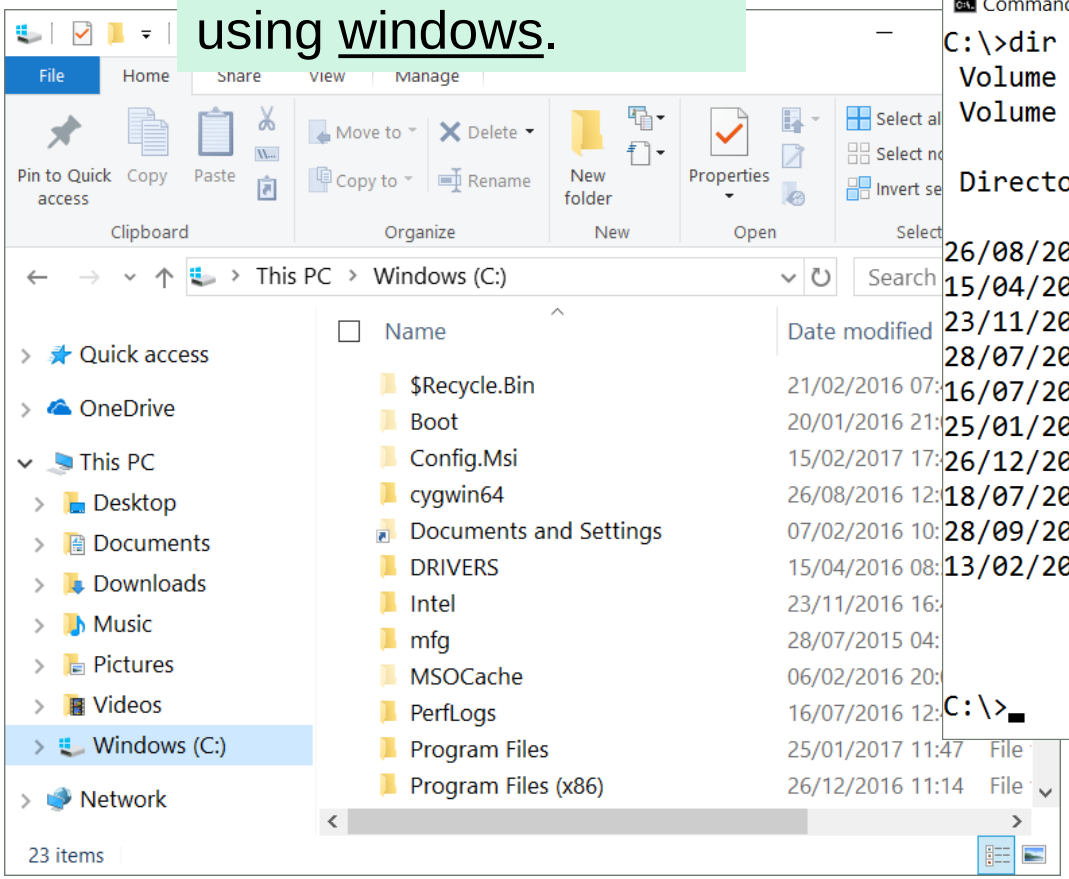
If you type **`javac`** on the command line and you get the message below, then something has gone wrong with your installation of the JDK. Try again!

```
'javac' is not recognized as an internal or
external command, operable program or batch file.
```

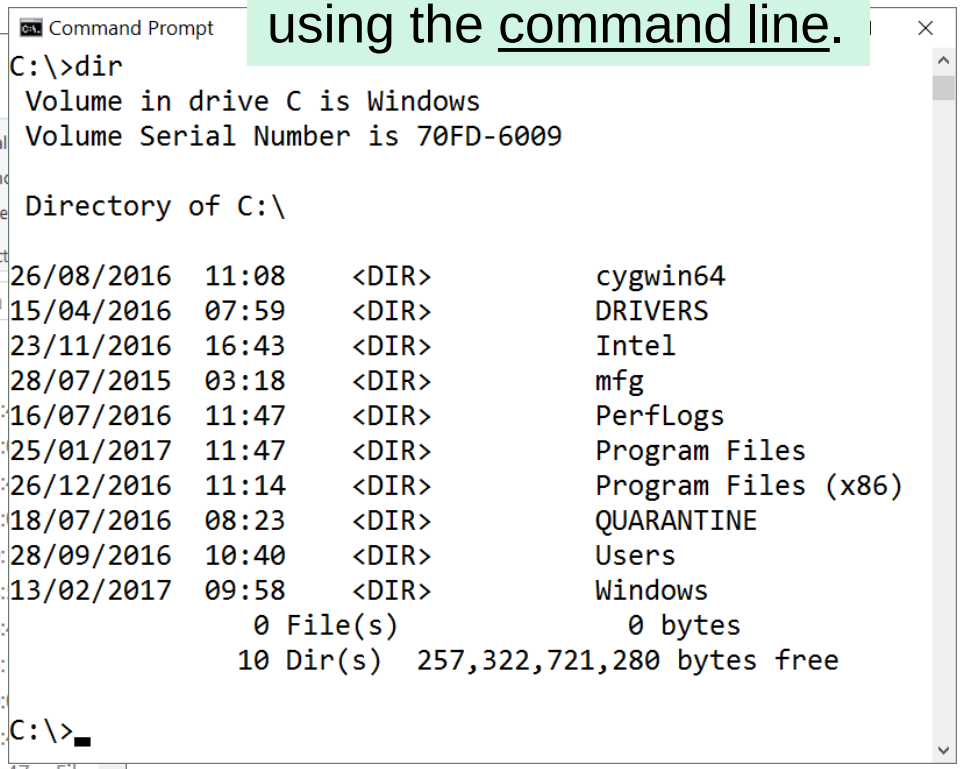
Essential Command Line Guide

- **Command Line** is the way in which computers used to run programs, i.e. without the use of windows!
- Using the command line, we can do many file management operations.

Listing files on a drive, using windows.



Listing files on a drive, using the command line.



Navigation & File Matching Metacharacters

`.` (dot) means ‘the current directory’

`..` (dot dot) means ‘the parent directory’

Note: The directory one level above the ‘current’ directory is the ‘parent’ directory, so the command `cd ..` will move you up one level.

- Operations on multiple files and be achieved by using the wildcard character `*`.
- When using `*`, DOS (the `cmd` window) will pattern-match the `*` portion.
- The character `?` Matches any character.

Useful `cmd` window commands (Windows OS computers)

CD	Displays the name of or changes the current directory.
CLS	Clears the screen.
CMD	Starts a new instance of the Windows command interpreter.
COPY	Copies one or more files to another location.
DATE	Displays or sets the date.
DEL	Deletes one or more files.
DIR	Displays a list of files and subdirectories in a directory.
EXIT	Quits the CMD.EXE program (command interpreter).
HELP	Provides Help information for Windows commands.
MKDIR	Creates a directory.
MOVE	Moves one or more files from one directory to another directory.
PATH	Displays or sets a search path for executable files.
RENAME	Renames a file or files.
RMDIR	Removes a directory.
SET	Displays, sets, or removes Windows environment variables.
TIME	Displays or sets the system time.

For more information on a specific command, type **`HELP command-name`**.

Examples of cmd window commands (1/2)

Command Prompt

```
C:\Teaching\EBU4201>dir *.xls
 Volume in drive C is Windows
 Volume Serial Number is 70FD-6009

 Directory of C:\Teaching\EBU4201

16/02/2017  09:05             9,446 cwinfodee0e4ff.xlsx
16/02/2017  09:06            10,451 cwinfodee0e47f.xlsx
16/02/2017  09:06            12,587 cwinfodee0e49f.xlsx
16/02/2017  09:07            14,647 cwinfodee0e51f.xlsx
16/02/2017  09:05             7,843 cwinfodee0e4bf.xlsx
16/02/2017  09:05             7,869 cwinfodee0e4df.xlsx
16/02/2017  09:08            17,846 examCN.xlsx
16/02/2017  09:08            14,157 LabAttendance2015.16.xlsx
               8 File(s)         94,846 bytes
               0 Dir(s)  257,279,377,408 bytes free

C:\Teaching\EBU4201>
```

Command Prompt

```
C:\Teaching\EBU4201>dir cw*.xls
 Volume in drive C is Windows
 Volume Serial Number is 70FD-6009

 Directory of C:\Teaching\EBU4201

16/02/2017  09:05             9,446 cwinfodee0e4ff.xlsx
16/02/2017  09:06            10,451 cwinfodee0e47f.xlsx
16/02/2017  09:06            12,587 cwinfodee0e49f.xlsx
16/02/2017  09:07            14,647 cwinfodee0e51f.xlsx
16/02/2017  09:05             7,843 cwinfodee0e4bf.xlsx
16/02/2017  09:05             7,869 cwinfodee0e4df.xlsx
               6 File(s)         62,843 bytes
               0 Dir(s)  257,279,377,408 bytes free

C:\Teaching\EBU4201>
```

Examples of `cmd` window commands (2/2)

Command Prompt

```
C:\java\EBU4201_2016.17>dir *First*Program.*
 Volume in drive C is Windows
 Volume Serial Number is 70FD-6009

 Directory of C:\java\EBU4201_2016.17

16/02/2017  09:21               566 FirstProgram.java
16/02/2017  09:18             3,725 MyFirstJavaProgram.class
16/02/2017  09:17               938 MyFirstJavaProgram.java
               3 File(s)          5,229 bytes
               0 Dir(s)  257,279,373,312 bytes free

C:\java\EBU4201_2016.17>dir ??First*Program.*
 Volume in drive C is Windows
 Volume Serial Number is 70FD-6009

 Directory of C:\java\EBU4201_2016.17

16/02/2017  09:18             3,725 MyFirstJavaProgram.class
16/02/2017  09:17               938 MyFirstJavaProgram.java
               2 File(s)          4,663 bytes
               0 Dir(s)  257,279,373,312 bytes free

C:\java\EBU4201_2016.17>
```

Command Prompt

```
C:\java\EBU4201_2016.17>dir
 Volume in drive C is Windows
 Volume Serial Number is 70FD-6009

 Directory of C:\java\EBU4201_2016.17

16/02/2017  09:16    <DIR>          .
16/02/2017  09:16    <DIR>          ..
16/02/2017  09:21               566 FirstProgram.java
16/02/2017  09:18             3,725 MyFirstJavaProgram.class
16/02/2017  09:17               938 MyFirstJavaProgram.java
               3 File(s)          5,229 bytes
               2 Dir(s)  257,279,242,240 bytes free

C:\java\EBU4201_2016.17>
```

Basic Command Line: Windows and Unix (*) computers

Not all lecturers use a Windows computer; this slide is intended to help students to follow exercises that may be demonstrated with either a Linux or Mac OS computer.

IMPORTANT:

1. This should be used together with information in QMplus, under **Teaching Block 1 > Useful Resources**.
2. All Java specific commands (e.g. **`javac`**, **`java`**, **`javadoc`**) are the same across different Operating Systems.
3. All navigation commands are given here in their simplified form only; commands may have other options that can be (at times) useful.

(*) This is applicable to both Linux and Mac OS computers.

(**) Be careful when using this command; there is no 'un-delete' option!

(***) This command is most useful with text files (e.g. those with a **`.txt`** or **`.java`** extension).

Windows command	Brief Description	Unix command
`dir <dirName>`	List information about files (in directory **`dirName`**, if specified)	`ls -l <dirName>`
`cd dirName`	Change to directory **`dirName`**; move to parent directory with **`'..'`**	`cd dirName`
`mkdir dirName`	Create a directory (i.e. a collection of files) called **`dirName`**, if it does not already exist	`mkdir dirName`
`del filename`	Remove/delete file called **`filename`**	`rm filename` (**)
`del dirName`	Remove/delete directory called **`dirName`**	`rmdir dirName` (**)
`copy <f> <dir>`	Copy file **`f`** to directory **`dir`**	`cp <f> <dir>`
`type <fileName>` (***)	Display the contents of file **`fileName`** on the standard output stream (i.e. usually the screen)	`cat <fileName>` (**)
`ren <f1> <f2>`	Rename file **`f1`** to file called **`f2`**	`mv <f1> <f2>`
(upper arrow key) ↑	Repeat previous command; very useful!	(upper arrow key) ↑
`exit`	Exit from the command line window; you can also exit by clicking the "X" button on the top right corner of the window.	`exit` (*or* `logout`)